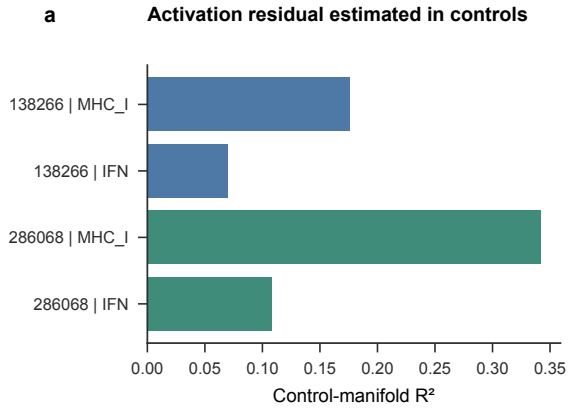
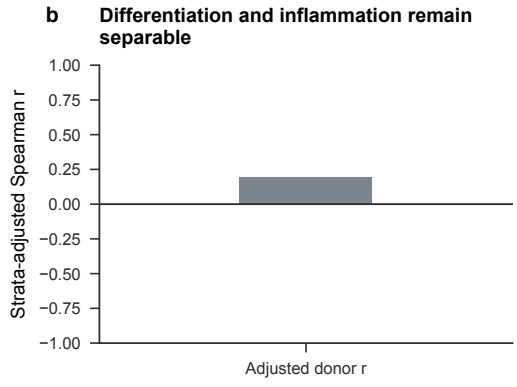
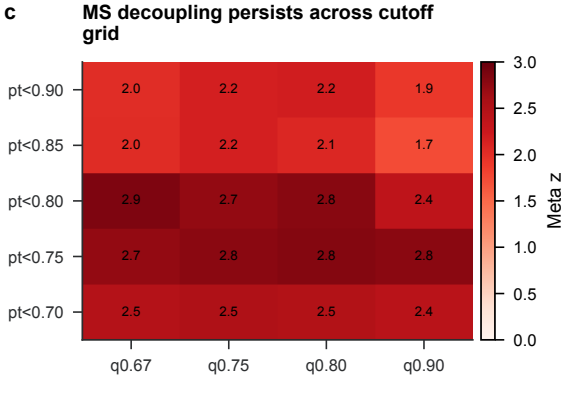
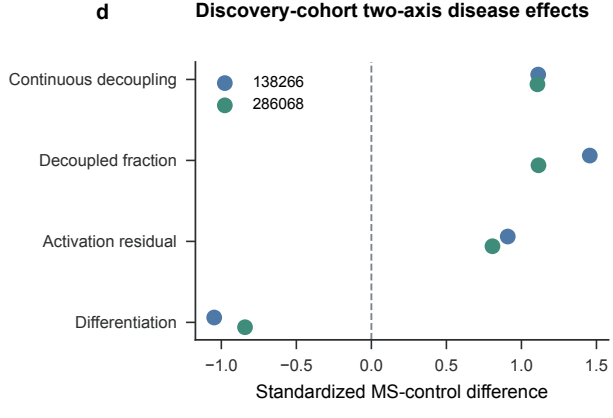

a
Activation residual estimated in controls
138266 | MHC_I
138266 | IFN
286068 | MHC_I
286068 | IFN
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Control-manifold R²


b
Differentiation and inflammation remain separable
Strata-adjusted Spearman r
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Adjusted donor r


c
MS decoupling persists across cutoff grid
pt<0.90
pt<0.85
pt<0.80
pt<0.75
pt<0.70
2.0
2.2
2.2
1.9
2.0
2.2
2.1
1.7
2.9
2.7
2.8
2.4
2.7
2.8
2.8
2.8
2.5
2.5
2.5
2.4
q0.67
q0.75
q0.80
q0.90
Meta z
3.0
2.5
2.0
1.5
1.0
0.5
0.0


d
Discovery-cohort two-axis disease effects
Continuous decoupling
Decoupled fraction
Activation residual
Differentiation
138266
286068
−1.0
−0.5
0.0
0.5
1.0
1.5
Standardized MS-control difference